

Agile Wavelength-Switchable Self-Injection-Locked Narrow Linewidth DBR Laser for Coherent Wavelength Routing

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Abstract We propose a hybrid self-injection-locked DBR laser with a Si_3N_4 micro-ring external cavity for coherent wavelength routing. Enabled by optical feedback over C-band ITU channels, it achieves 89 ns switching, stable wavelength locking, and a 2.5 kHz linewidth for high-speed optical switching. ©2026 The Author(s)

Introduction

Driven by the evolution of dynamic optical switching networks, reconfigurable optical interconnects, and inter-satellite wavelength-routed laser communications, the development of agile wavelength-switchable laser sources has become an urgent necessity [1, 2]. In optical switching architectures combining arrayed waveguide grating routers (AWGRs) and fast tunable lasers, the signal laser selects the destination port by rapidly switching its operating wavelength. Consequently, the tuning speed directly determines the switching latency and the granularity of traffic scheduling [3-5]. Rapid wavelength switching is highly advantageous as it enables low-latency, fine-grained optical burst switching to effectively manage bursty data traffic. However, emerging scenarios such as inter-satellite routing and coherent optical communications impose even stricter requirements. To support long-distance, high-capacity coherent detection links, the laser source must deliver not only sub-microsecond switching agility but also an ultra-narrow linewidth and excellent frequency stability [6]. Recently, significant progress has been made in monolithic and hybrid integrated tunable lasers [7-10]. Monolithically integrated multi-wavelength distributed feedback (DFB) laser arrays can achieve nanosecond-scale switching speeds. However, limited by the intrinsic loss of active III-V materials, their linewidths are typically confined to the megahertz range, which severely restricts their application in high-order coherent systems. Conversely, widely tunable external-cavity lasers can achieve narrow linewidths, but they usually suffer from slow tuning responses and are susceptible to transient frequency drifting or mode hopping during dynamic transitions. Therefore, a fundamental trade-off exists between switching agility and linewidth. Consequently, there is a compelling demand for an integrated laser source that can simultaneously provide agile wavelength switching,

reliable wave-length locking, and kilohertz-level linewidth for standard ITU channel configurations.

In this paper, we propose and experimentally demonstrate an agile wavelength-switchable, narrow-linewidth laser tailored for coherent wavelength routing. The device is based on a hybrid integration architecture, combining a distributed Bragg reflector (DBR) laser chip and an Si_3N_4 external cavity. The DBR chip provides rapid electrical tuning, while the high-Q micro-ring external cavity offers narrowband optical feedback. Through robust self-injection-locking (SIL), the lasing frequency is instantaneously captured and stabilized, which effectively suppresses transient frequency drifts and significantly narrows the linewidth. Operating over C-band 50-GHz-spaced ITU channels, the proposed device overcomes traditional performance limitations, enabling an unprecedented combination of agile switching, instantaneous wavelength locking, and extreme linewidth narrowing. Specifically, it simultaneously achieves a fast switching time of 89 ns and an ultra-narrow linewidth of 2.5 kHz. These results verify the immense potential of this hybrid integrated laser source for high-speed coherent optical switching and inter-satellite routing applications.

Device Design and Experimental Setup

The proposed agile wavelength-switchable laser source employs a hybrid-integrated architecture that couples an active DBR chip with a passive Si_3N_4 external-cavity chip via dual-lens coupling, as depicted in Fig. 1(a). This architecture leverages the dynamic interplay between active wavelength tuning and passive wavelength locking. Specifically, the III-V DBR chip functions as a high-speed tuning engine. As shown in Fig. 1(c), the modulated grating Y-branch DBR chip consists of left and right reflectors, a phase section, a gain section, and a semiconductor optical amplifier (SOA) section, utilizing carrier density

modulation to drive nanosecond-scale wavelength switching. Concurrently, the Si_3N_4 external cavity operates as a high-Q narrowband optical filter, providing selective optical feedback for simultaneous linewidth reduction and instantaneous, all-optical wavelength stabilization. Ultimately, this synergistic mechanism of active tuning and passive feedback locking is essential for achieving sub-100-ns switching dynamics while preserving a kilohertz-level emission linewidth.

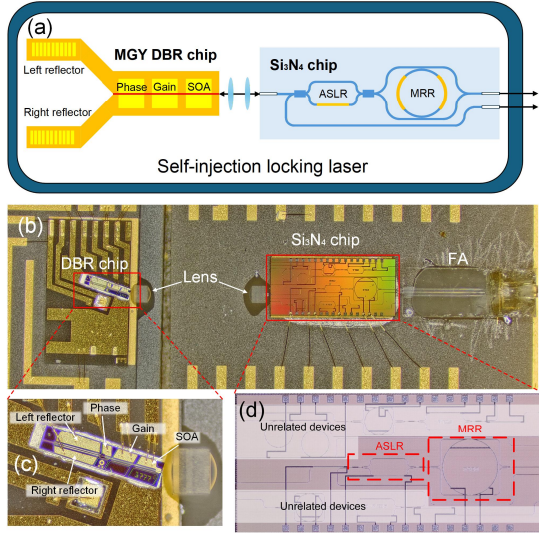


Fig. 1: (a) Internal structural diagram and (b) micrograph of the hybrid-integrated DBR/ASLR- Si_3N_4 SIL laser, (c) Microscopic image of the DBR chip, (d) Microscope photo of the external cavity chip used for SIL laser.

As detailed in Fig. 1(d), the Si_3N_4 external cavity chip features an adjustable Sagnac loop reflector (ASLR) and a micro-ring resonator (MRR). The ASLR serves as more than a simple feedback loop, its phase and splitting ratio can be precisely controlled through thermo-optic tuning. This allows for the optimization of the feedback phase and strength, thereby enlarging the effective capture range of the self-injection-locking process. The MRR, with a loaded Q factor of 1.16×10^6 , is designed with a 50 GHz free spectral range (FSR) to match standard ITU channels, and its resonances can be accurately aligned through temperature control. This design fulfils the stringent requirements of next-generation optical routing systems.

The experimental setup for evaluating the dynamic wavelength-routing performance of the proposed laser is illustrated in Fig. 2. High-speed driving signals generated by an arbitrary waveform generator (AWG) are applied to the tuning sections of the MGY-DBR chip to facilitate agile wavelength switching. In this configuration, the ASLR is maintained at optimized static conditions to ensure stable feedback and robust SIL across different operating states. The laser output is directed through an optical isolator and

subsequently routed into an AWGR. This allows for the spatial separation of distinct wavelength channels, which are then detected by an array of high-speed photodetectors (PDs) and recorded by a real-time oscilloscope (OSC) to comprehensively analyze the switching latency and dynamic stability.

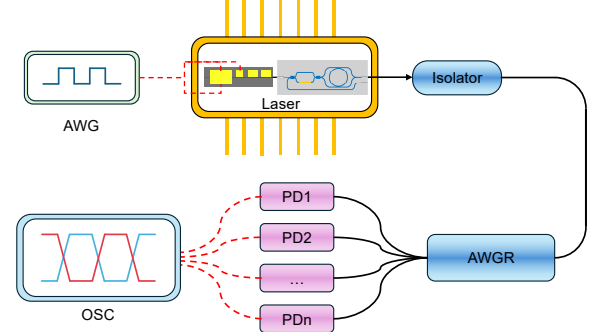


Fig. 2: Diagram of the experimental setup for dynamic wavelength routing and switching.

Experimental results

The static spectral performance of the proposed SIL laser was characterized under robust locking conditions, as shown in Fig. 3. Stable single-mode lasing was obtained at multiple representative wavelengths, verifying the broadband tunability of the device. While Fig. 3 illustrates discrete examples, the device is intrinsically capable of full C-band operation aligned to the 50-GHz-spaced ITU channels. Under the present hybrid-integration conditions, the measured fiber-coupled output power is approximately 0.41 mW, with a side-mode suppression ratio (SMSR) exceeding 41 dB.

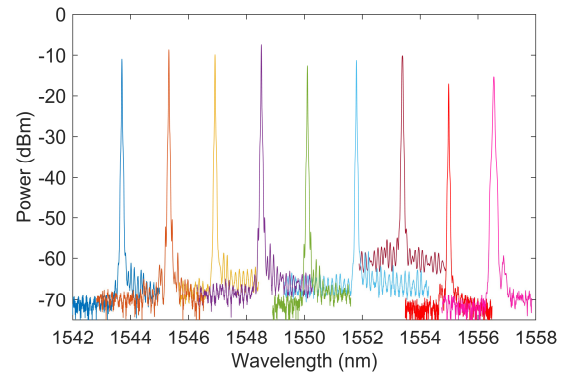


Fig. 3: Representative output spectra across the C-band.

The frequency-noise characteristics of the device were further investigated, as shown in Fig. 4, indicating that the selective feedback provided by the Si_3N_4 external cavity effectively suppresses laser phase fluctuations. Based on the measured frequency-noise spectrum, the intrinsic laser linewidth is determined to be approximately 2.5 kHz, which is rigorously extracted from the white-noise floor level (h_0) using the relation $\Delta\nu = \pi h_0$. This result confirms that the proposed architecture provides exceptional phase-noise suppression,

thereby achieving an ultra-narrow linewidth perfectly suited for high-capacity coherent detection applications.

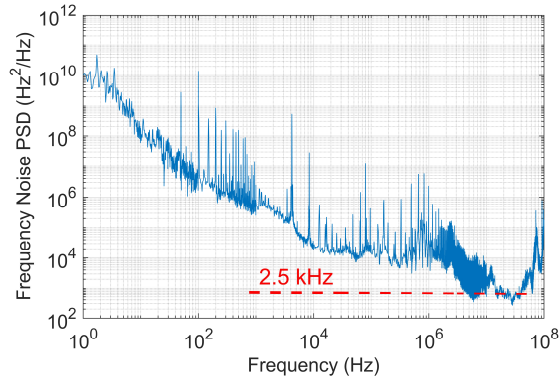


Fig. 4: The laser frequency noise PSD.

We systematically investigated the dynamic wavelength-routing capability of the SIL laser. Complementary electrical signals generated by a high-speed AWG were applied to the tuning sections of the DBR chip to enable rapid wavelength switching. As shown in Fig. 5(a), stable and highly repeatable periodic switching between 1546.92 nm and 1554.94 nm was achieved. Notably, over multiple consecutive switching cycles, the lasing wavelength remained precisely locked to the ITU-grid without requiring any additional external wavelength-feedback control loop. An enlarged view of a single switching transient is presented in Fig. 5(b). Evaluated using the standard 10%-90% criterion, the measured typical switching time for the rising edge is 89 ns. Crucially, within a window of less than 100 nanoseconds, the laser not only performs broadband frequency hopping but also instantaneously establishes SIL to the target channel without exhibiting observable long-term frequency oscillations. At present, the measured switching speed is mainly limited by the bandwidth of the external driving circuitry and PCB packaging. Given that the SIL establishment is nearly instantaneous, the fundamental temporal limit of the overall switching and locking dynamics is dictated solely by the intrinsic carrier recombination lifetime of the DBR tuning sections. Consequently, the switching speed holds immense potential to be further pushed into the nanosecond regime through optimized high-frequency packaging.

Although the current proof-of-concept utilizes dual-lens-coupled hybrid integration, the proposed architecture offers inherent scalability. This design is amenable to direct migration onto 300mm heterogeneous silicon photonics platforms. Such wafer-scale integration would effectively mitigate facet coupling losses, thereby elevating the output power to the tens-of-milliwatts regime to address the stringent link budgets of

high-capacity coherent optical networks.

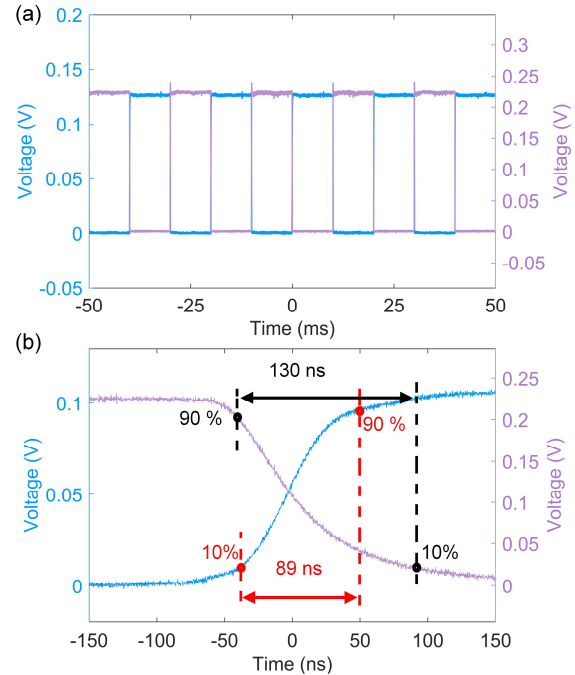


Fig. 5: (a) Periodic switching waveforms between two wavelength channels, (b) Enlarged view of a single switching event, with a measured rise time of 89 ns and a fall time of 130 ns.

Conclusions

In conclusion, we demonstrate an agile wavelength-switchable, narrow-linewidth laser tailored for coherent wavelength routing. By synergistically combining an active MGY-DBR chip with an ASLR-Si₃N₄ external cavity, the proposed hybrid architecture enables simultaneous rapid tuning, instantaneous all-optical stabilization, and extreme linewidth suppression. Experimental results show an ultra-narrow linewidth of 2.5 kHz. Notably, the device achieves a remarkable 89 ns switching time across 50-GHz-spaced ITU channels without observable frequency ringing. This work highlights the immense potential of the architecture for next-generation coherent optical switching and inter-satellite wavelength-routing applications.

Acknowledgements

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